```
#include <iostream>
using namespace std;

#define INF 99999

int main()
{
    int n;

    cout << "Enter number of offices: ";
    cin >> n;

    if (n < 1 || n > 20)
    {
        cout << "Invalid number of offices.";
        return 0;
    }

    int cost[20][20];

    cout << "Enter cost matrix (0 for no link):\n";

    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cin >> cost[i][j];
```

```
        if (i != j && cost[i][j] == 0)
            cost[i][j] = INF;
    }
}

int visited[20] = {0};
int edges = 0;
int totalCost = 0;

visited[0] = 1;

cout << "\nSelected fiber-optic links:\n";

while (edges < n - 1)
{
    int minCost = INF;
    int u = -1, v = -1;

    for (int i = 0; i < n; i++)
    {
        if (visited[i])
        {
            for (int j = 0; j < n; j++)
            {
                if (!visited[j] &&
                    cost[i][j] < minCost)
```

```
            {
                minCost = cost[i][j];
                u = i;
                v = j;
            }
        }
    }
}

if (u == -1)
{
    cout << "Network cannot connect all offices.\n";
    return 0;
}

visited[v] = 1;
edges++;
totalCost += minCost;

cout << "Office " << u + 1
     << " - Office " << v + 1
     << " : Cost = " << minCost
     << ", Running Total = "
     << totalCost << endl;
}

cout << "\nMinimum total cable cost = "
```

```
        << totalCost << endl;


    return 0;
}
```
Sample Input